

# EE3621: Digital Signal Processing

## Lecture 28: Bilinear Transformation (BLT) Method

(III B.Tech EEE)

September 9, 2026

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## 1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Derive** the Bilinear Transformation algebraic mapping and frequency warping relation.
  2. **Prewarp** digital filter specifications into continuous analog frequencies.
  3. **Design** Butterworth and Chebyshev IIR digital filters using the Bilinear Transformation.
  4. **Compare** the Bilinear Transformation with the Impulse Invariance method.
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## 2. MATHEMATICAL FOUNDATIONS

### 2.1 Derivation via Trapezoidal Integration

Consider the first-order differential equation  $\frac{dy(t)}{dt} = x(t)$ . Integrating over  $[(n-1)T_d, nT_d]$  using the trapezoidal rule:

$$y(nT_d) - y((n-1)T_d) = \int_{(n-1)T_d}^{nT_d} x(t)dt \approx \frac{T_d}{2} [x(nT_d) + x((n-1)T_d)]$$

In discrete notation:

$$y[n] - y[n-1] = \frac{T_d}{2} [x[n] + x[n-1]]$$

Taking the Z-Transform:

$$Y(z)(1 - z^{-1}) = \frac{T_d}{2} X(z)(1 + z^{-1}) \implies H(z) = \frac{Y(z)}{X(z)} = \frac{T_d}{2} \frac{1 + z^{-1}}{1 - z^{-1}}$$

Since the analog integrator transfer function is  $H_a(s) = \frac{1}{s}$ , equating  $H_a(s) = H(z)$  yields:

$$s = \frac{2}{T_d} \frac{1 - z^{-1}}{1 + z^{-1}} = \frac{2}{T_d} \frac{z - 1}{z + 1}$$

### 2.2 Frequency Warping and Prewarping

Substitute  $s = j\Omega$  and  $z = e^{j\omega}$ :

$$j\Omega = \frac{2}{T_d} \frac{1 - e^{-j\omega}}{1 + e^{-j\omega}} = \frac{2}{T_d} \frac{e^{j\omega/2} - e^{-j\omega/2}}{e^{j\omega/2} + e^{-j\omega/2}} = j \frac{2}{T_d} \tan\left(\frac{\omega}{2}\right)$$

$$\Omega = \frac{2}{T_d} \tan\left(\frac{\omega}{2}\right)$$

- Prewarping Formulas:

$$\Omega_p = \frac{2}{T_d} \tan\left(\frac{\omega_p}{2}\right), \quad \Omega_s = \frac{2}{T_d} \tan\left(\frac{\omega_s}{2}\right)$$

## Pictorial Representations

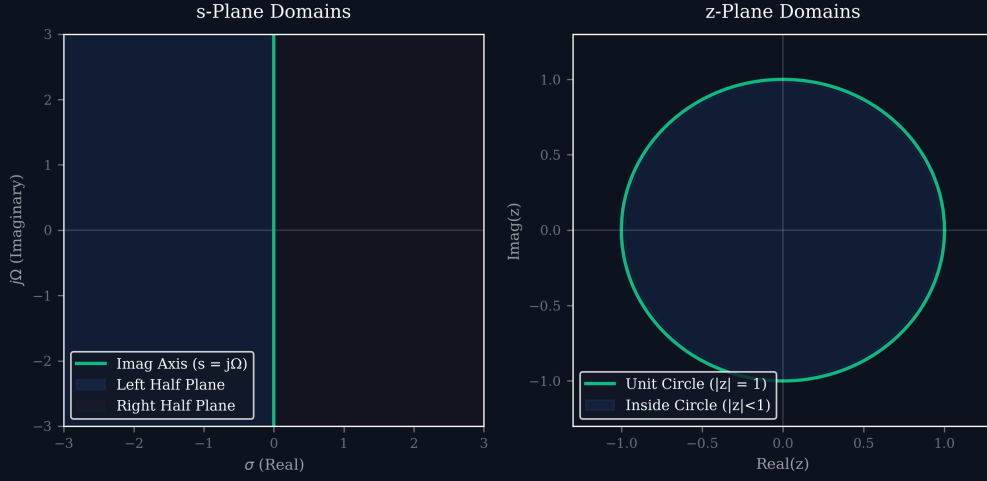


Figure 1: Bilinear S To Z

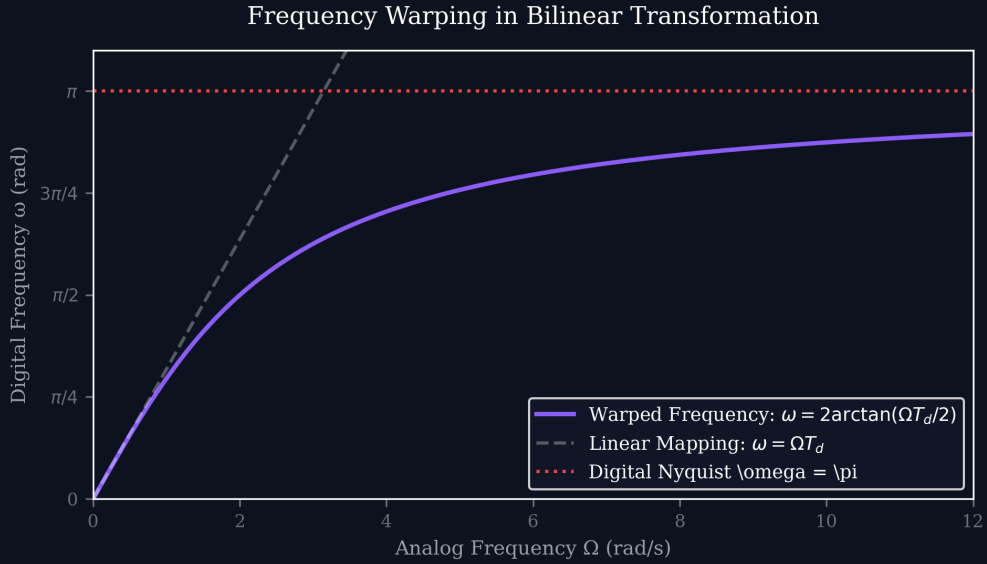


Figure 2: Bilinear Frequency Warping

### 3. WORKED NUMERICAL EXAMPLES

#### Example 28.1: 1st-Order Lowpass Filter via BLT

**Problem:** Design a digital lowpass filter with a 3-dB cutoff frequency  $\omega_c = 0.2\pi$  rad/sample from an analog prototype  $H_a(s) = \frac{\Omega_c}{s + \Omega_c}$  using the Bilinear Transformation ( $T_d = 1$  s).

**Solution:**

##### 1. Prewarp the Cutoff Frequency:

$$\Omega_c = \frac{2}{T_d} \tan\left(\frac{\omega_c}{2}\right) = \frac{2}{1} \tan(0.1\pi) = 2 \times 0.3249 = 0.6498 \text{ rad/s}$$

##### 1. Analog Transfer Function:

$$H_a(s) = \frac{0.6498}{s + 0.6498}$$

##### 1. Apply Bilinear Transformation ( $s = 2\frac{1-z^{-1}}{1+z^{-1}}$ ):

$$H(z) = \frac{0.6498}{s + 0.6498} \Big|_{s=2\frac{1-z^{-1}}{1+z^{-1}}} = \frac{0.6498}{2\frac{1-z^{-1}}{1+z^{-1}} + 0.6498} = \frac{0.6498(1+z^{-1})}{2(1-z^{-1}) + 0.6498(1+z^{-1})}$$

$$H(z) = \frac{0.6498 + 0.6498z^{-1}}{(2 + 0.6498) + (-2 + 0.6498)z^{-1}} = \frac{0.6498(1+z^{-1})}{2.6498 - 1.3502z^{-1}} = \frac{0.2452(1+z^{-1})}{1 - 0.5095z^{-1}}$$

##### 1. Verification:

\* At DC ( $\omega = 0, z = 1$ ):  $H(1) = \frac{0.2452(2)}{1-0.5095} = \frac{0.4904}{0.4905} \approx 1.0$  (0 dB). \* At half-sampling ( $\omega = \pi, z = -1$ ):  $H(-1) = \frac{0.2452(0)}{1+0.5095} = 0$  ( $-\infty$  dB).

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### 4. UNIVERSITY EXAMINATION QUESTIONS & MARKING RUBRIC

#### Question 1 (15 Marks)

(a) Derive the Bilinear Transformation and the frequency warping equation. Explain the necessity of prewarping. \*(8 Marks)\* (b) Design a digital Butterworth lowpass filter satisfying:

- $0.8 \leq |H(e^{j\omega})| \leq 1.0$  for  $0 \leq \omega \leq 0.2\pi$
- $|H(e^{j\omega})| \leq 0.2$  for  $0.6\pi \leq \omega \leq \pi$

Use Bilinear Transformation with  $T_d = 1$  s. \*(7 Marks)\*

**Model Answer & Step-by-Step Marking Rubric:**

##### • Part (a):

\* Trapezoidal derivation of  $s = \frac{2}{T_d} \frac{1-z^{-1}}{1+z^{-1}}$  \*(4 Marks)\* \* Warping derivation  $\Omega = \frac{2}{T_d} \tan(\omega/2)$  and prewarping explanation \*(4 Marks)\*

##### • Part (b):

\* Prewarp:  $\Omega_p = 2 \tan(0.1\pi) = 0.6498$ ,  $\Omega_s = 2 \tan(0.3\pi) = 2.7528$  \*(2 Marks)\* \* Specs:  $A_p = -20 \log_{10}(0.8) = 1.9382$  dB,  $A_s = -20 \log_{10}(0.2) = 13.9794$  dB \*(1 Mark)\* \* Order calculation:  $N \geq \frac{\log_{10}[(10^{1.398}-1)/(10^{0.1938}-1)]}{2 \log_{10}(2.7528/0.6498)} = \frac{\log_{10}(24/0.5623)}{2 \log_{10}(4.236)} = \frac{1.6303}{1.254} = 1.30 \implies \mathbf{N = 2}$  \*(2 Marks)\* \* Synthesize 2nd-order Butterworth  $H_a(s)$  and substitute  $s \rightarrow 2\frac{1-z^{-1}}{1+z^{-1}}$  \*(2 Marks)\*

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## 5. PYTHON VERIFICATION SCRIPT

```
import scipy.signal as signal
```

```
\title{\textbf{EE3621: Digital Signal Processing}} \Large Butterworth BLT design \Large  
b, a = signal.butter(2, 0.2, btype='low', analog=False)  
print("Digital Filter b:", np.round(b, 4))  
print("Digital Filter a:", np.round(a, 4))
```